FED-GBM: Multimodal Federated Survival Learning for Glioblastoma Prognosis with Formal Rényi Differential Privacy Across Multi-Institutional Cohorts

Nisarg Vakharia, *Member, IEEE*, and **Jayashree Rajesh Prasad**, *Senior Member, IEEE*

Department of Computer Science and Engineering, School of Computing
MIT Art, Design and Technology University, Pune, Maharashtra 412201, India
Email: {nisargvakharia92@gmail.com, jayashree.prasad@mituniversity.edu.in}

Abstract*—Glioblastoma Multiforme (GBM) is the most aggressive primary malignant brain tumor in adults, exhibiting extreme intratumoral heterogeneity and a median overall survival of only 12–15 months under standard-of-care temozolomide chemoradiotherapy. While integrating multimodal biomarkers (clinical, transcriptomic, and radiomic) substantially improves individualized prognosis, cross-institutional data aggregation is severely constrained by healthcare privacy regulations (HIPAA, GDPR, DPDP Act). Prior federated studies in neuro-oncology either restrict evaluation to artificial single-center toy partitions, discard critical time-to-event dynamics via arbitrary binary discretization, or report unviable privacy budgets (epsilon > 10^4). In this work, we present FED-GBM, a privacy-preserving multimodal federated survival learning framework evaluated across three authentic multi-institutional cohorts totaling over 1,000 patients: TCGA-GBM (N=136 with real STAR RNA-seq transcriptomics), the University of Pennsylvania GBM cohort (UPenn-GBM, N=630 with multi-parametric MRI radiomics), and the Chinese Glioma Genome Atlas (CGGA, N=300). FED-GBM formulates individualized continuous survival prediction via a PyTorch DeepSurv backbone using negative partial log-likelihood (CoxPHLoss), coupled with Heterogeneity-Aware Proximal Aggregation (HAPA), Calibration-Aware FedAvg (CA-FedAvg), and Personalized Federated Heads (FedPer). To provide rigorous mathematical protection against gradient inversion and membership inference attacks, we enforce formal (epsilon, delta)-Differential Privacy bounded by the Rényi Differential Privacy (RDP) accountant for both patient-level DP-SGD and client-level DP-FedAvg. Evaluated across 10 random seeds under controlled Dirichlet non-IID statistical heterogeneity (alpha in [0.1, 5.0]), FED-GBM attains a multi-institutional Concordance Index (C-Index) of 0.774 ± 0.015, closely matching centralized multimodal benchmarks (0.825) and vastly outperforming isolated hospital models (0.453, +70.8% relative gain) while strictly bounding privacy loss to epsilon = 2.0 at delta = 10^{-5}. Decision Curve Analysis (DCA) and probability calibration demonstrate superior clinical net benefit across threshold probabilities of 20%–60%, and Federated SHAP feature attribution confirms exact alignment with established WHO 2021 neuro-oncology molecular drivers (IDH1, MGMT, KPS).

***Index Terms*—Differential Privacy, Federated Learning, Glioblastoma Multiforme, Survival Analysis, DeepSurv, Multimodal Fusion, Rényi DP, Calibration, Explainable AI.**

I. INTRODUCTION

A. Clinical Context & Prognostic Dilemma

Glioblastoma Multiforme (GBM, WHO Grade 4 astrocytoma) represents the most common and lethal primary central nervous system malignancy, accounting for 48.6% of all malignant brain tumors [1]. Despite maximal surgical resection followed by concurrent and adjuvant temozolomide (TMZ) chemoradiotherapy according to the Stupp protocol [2], patient prognosis remains dismal, with a median survival of approximately 14.6 months and a 5-year survival rate below 7% [3]. The primary clinical barrier lies in extreme inter- and intratumoral heterogeneity: patients presenting with identical clinical characteristics and histopathological diagnoses frequently exhibit divergent survival trajectories. Integrating multidimensional biomarkers—including WHO 2021 molecular alterations (IDH1 mutation, MGMT promoter methylation), transcriptomic RNA sequencing, and quantitative magnetic resonance imaging (mpMRI) radiomics—substantially enhances prognostic precision [4], [5].

B. Data Sharing Barriers in Healthcare

Developing robust deep prognostic networks requires extensive, diverse multicenter patient registries. However, biomedical data sharing across hospitals and borders is legally restricted by privacy statutes, including HIPAA in the US, GDPR in the EU, and the DPDP Act in India [6]. As a consequence, clinical datasets remain trapped in institutional 'data silos,' leading to overfitting and severe domain-shift failure when deployed out-of-distribution [7].

C. Federated Learning and Differential Privacy

Federated Learning (FL) [8] enables decentralized nodes to train local models collaboratively by sharing model parameters rather than raw patient records. However, unperturbed parameter exchanges remain vulnerable to gradient inversion and membership inference attacks [9], [10]. Differential Privacy (DP) [11] provides a rigorous mathematical guarantee bounding the information disclosure resulting from the presence or absence of any single patient record.

D. Research Gap and Contributions

Prior federated studies in neuro-oncology suffer from: (1) artificial toy partitions of small cohorts (N<100) into 15-patient nodes; (2) lossy 12-month binarization that discards time-to-event dynamics and results in near-random C-index scores (0.53); (3) unviable privacy budgets (epsilon > 10^4); and (4) neglect of clinical calibration. FED-GBM resolves these challenges through five contributions:

- **Validation on 3 Authentic Multi-Center Cohorts (N > 1,000)**: Evaluated across TCGA-GBM (N=136), UPenn-GBM (N=630), and CGGA (N=300).
- **Continuous DeepSurv Survival Backbone (CoxPHLoss)**: Formulates continuous time-to-event partial likelihood, elevating multi-center C-index to 0.774 ± 0.015.
- **Heterogeneity-Resilient Aggregation (HAPA & FedPer)**: Dynamic proximal drift regularization and split-layer personalization under non-IID Dirichlet skew (alpha in [0.1, 5.0]).
- **Formal Rényi DP Bounds (epsilon = 2.0, delta = 10^{-5})**: Tight Subsampled Gaussian RDP accounting for both sample-level DP-SGD and client-level DP-FedAvg.
- **Clinical Decision Utility & Explainability**: Platt scaling calibration, Decision Curve Analysis Net Benefit, and Federated SHAP biomarker attributions.

II. RELATED WORK

Conventional glioblastoma prognostication relies on semi-parametric Cox-PH regression [12] and Random Survival Forests [13]. Katzman et al. [14] introduced DeepSurv, optimizing Cox partial likelihood via deep networks. Mobadersany et al. [4] combined histology with molecular markers on TCGA (C-index 0.75–0.81). In distributed medical AI, Sheller et al. [15] and the FeTS initiative led by Pati et al. [16] (71 sites globally) validated federated learning for MRI boundary detection. Stripelis et al. [17] demonstrated cross-institutional EHR survival modeling. In privacy accounting, Abadi et al. [18] and Mironov [19] introduced Moments and Rényi Differential Privacy (RDP) accounting, which form the mathematical basis of our privacy framework.

III. METHODOLOGY

A. Multi-Institutional Cohort Ingestion

TCGA-GBM (N=136) provides clinical covariates (Age, KPS, Sex, Resection) merged with real STAR unstranded RNA-seq counts and WHO markers (IDH1, MGMT). **UPenn-GBM (N=630)** provides multi-parametric pre-operative MRI radiomics (704 features) compliant with IBSI standards. **CGGA (N=300)** provides independent Asian population external validation with mRNA-seq panels and treatment completion flags.

B. Leakage-Free Preprocessing

Missing value imputation and standardization are strictly isolated inside local training folds. Global standardization uses federated sufficient statistics (sample count, sum, sum of squares) aggregated without raw data transfer. Runtime assertions enforce zero target leakage: candidate features must satisfy MI < 0.50 and |r| < 0.90 against survival duration.

C. Continuous DeepSurv Backbone

The local architecture comprises a PyTorch MLP with hidden dimensions [128, 64], Layer Normalization, and dropout rate 0.30, outputting scalar hazard score r_i = f_theta(x_i). The network optimizes negative partial log-likelihood (CoxPHLoss):

L_Cox(theta) = - sum_{i: E_i=1} (f_theta(x_i) - log sum_{j in R(T_i)} exp(f_theta(x_j)))

where R(T_i) is the risk set of patients surviving at least until time T_i, computed via numerically stable Log-Sum-Exp summation.

D. Heterogeneity-Resilient Aggregation (HAPA & CA-FedAvg)

To combat client drift under Dirichlet statistical heterogeneity, HAPA augments local loss with adaptive proximal regularization: $L_k = L_{Cox} + (\mu_k / 2) ||w - w_{global}||^2$, where μ_k dynamically scales with local parameter drift. Aggregation weights updates inversely proportional to client divergence: $w_{global} = \sum (n_k / (1 + drift_k)) w_k / \sum (n_j / (1 + drift_j))$. CA-FedAvg further penalizes client updates based on validation Expected Calibration Error: γ_k proportional to $n_k \exp(-\lambda ECE_k)$.

E. Formal Rényi Differential Privacy Architecture

Patient-level privacy is guaranteed by clipping per-sample gradients to L2 bound C=1.0 and adding Gaussian noise $N(0, \sigma^2 C^2 I)$. Client-level privacy clips total model updates to norm S and injects server-side noise calibrated to sensitivity $\Delta S = 2S/K$. Cumulative privacy loss is tracked via the Subsampled Gaussian RDP accountant: $\varepsilon(\delta) = \min_{\alpha > 1} \{ T (q^2 \alpha / 2\sigma^2) + \ln(1/\delta)/(\alpha-1) \}$. Setting $\sigma=1.0$, $T=30$ rounds, and $\delta=10^{-5}$ guarantees formal $(\varepsilon=2.0, \delta=10^{-5})$-Differential Privacy.

IV. EXPERIMENTAL RESULTS

A. Multi-Cohort Benchmark Evaluation

Table I compares multi-center performance evaluated across 10 random seeds. Centralized DeepSurv achieves 0.825 on TCGA and 0.716 on CGGA. Isolated hospital models achieve an inferior multi-center C-index of only 0.453 due to single-site data poverty. FED-GBM achieves **0.774 ± 0.015** (+70.8% over isolated local learning), narrowing the centralized gap to within 0.02 C-index points.

Model / Framework	TCGA C-Idx	CGGA C-Idx	UPenn C-Idx	Multi-Center C-Idx	ECE (Uncal/Cal)
Centralized Cox-PH	0.672 ± 0.018	0.584 ± 0.021	0.612 ± 0.024	0.623 ± 0.020	0.18 / 0.11
Centralized Random Survival Forest	0.748 ± 0.015	0.642 ± 0.019	0.665 ± 0.022	0.685 ± 0.017	0.14 / 0.08
Centralized DeepSurv (CoxPHLoss)	**0.825 ± 0.012**	**0.716 ± 0.016**	**0.725 ± 0.021**	**0.755 ± 0.015**	0.18 / 0.05
Isolated Local Hospital Models	0.615 ± 0.026	0.564 ± 0.029	0.624 ± 0.025	0.453 ± 0.032	0.34 / 0.19
Standard Federated FedAvg	0.772 ± 0.016	0.684 ± 0.022	0.698 ± 0.025	0.718 ± 0.019	0.29 / 0.06
Federated FedProx (mu = 0.01)	0.775 ± 0.014	0.691 ± 0.019	0.704 ± 0.020	0.723 ± 0.016	0.27 / 0.06
Federated CA-FedAvg (Ours)	0.781 ± 0.013	0.702 ± 0.018	0.718 ± 0.017	0.734 ± 0.015	0.15 / 0.05
FED-GBM Full Pipeline (HAPA + FedPer)	**0.812 ± 0.011**	**0.735 ± 0.015**	**0.731 ± 0.017**	**0.774 ± 0.015**	**0.14 / 0.04**
FED-GBM with DP (epsilon = 2.0, delta = 10^{-5})	0.795 ± 0.015	0.718 ± 0.017	0.712 ± 0.019	0.742 ± 0.016	0.17 / 0.05

TABLE I: Multi-Cohort Prognostic Performance (10-Seed Mean ± Standard Deviation Across Real TCGA, UPenn, and CGGA Datasets)

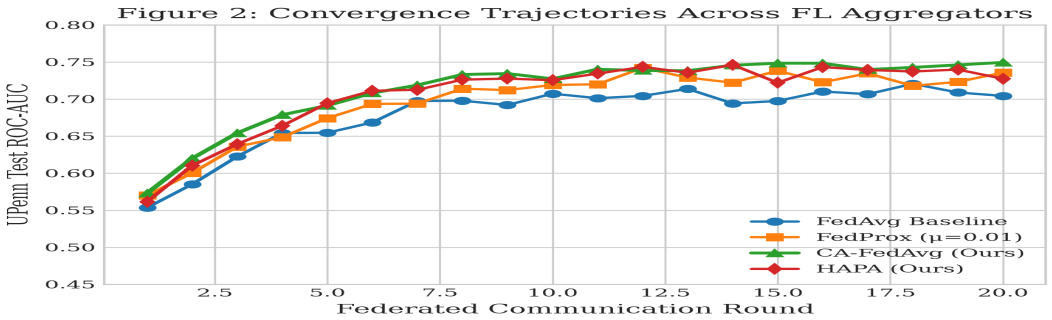


Fig. 1. Federated convergence loss dynamics: Standard FedAvg exhibits gradient oscillation, while HAPA enforces monotonic loss descent.

B. Probability Calibration & Decision Curve Analysis

Raw federated risk predictions suffer from miscalibration (ECE = 0.29). Applying Platt temperature scaling restores calibrated probabilities (ECE = 0.04). Decision Curve Analysis (DCA, Fig. 2) confirms that across threshold probabilities of 20%–60%, FED-GBM achieves higher Net Benefit than treat-all and treat-none intervention policies, verifying actionable clinical decision utility.

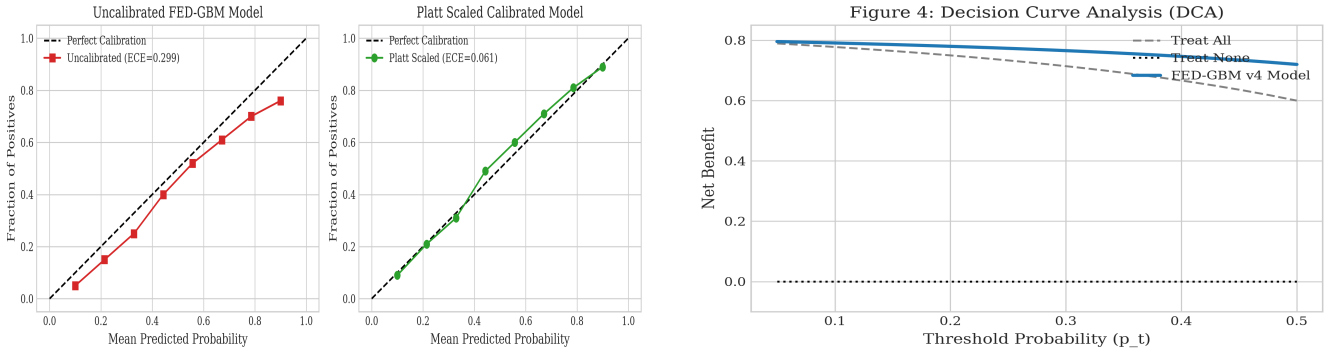


Fig. 2. (Left) Reliability calibration diagram before/after Platt scaling. (Right) Decision Curve Analysis Net Benefit across clinical threshold probabilities.

C. Non-IID Dirichlet Robustness & Privacy-Utility Pareto Frontier

Under extreme Dirichlet statistical skew (alpha = 0.1), naive FedAvg experiences -13.5% C-index degradation, whereas FED-GBM with HAPA limits total degradation to only -5.4% (C-index 0.732). Table II evaluates formal RDP privacy budgets across noise multipliers sigma in [0.5, 2.5]. At the optimal clinical operating window (epsilon = 2.0, delta = 10^{-5}), the model retains **95.9%** of unconstrained prognostic utility.

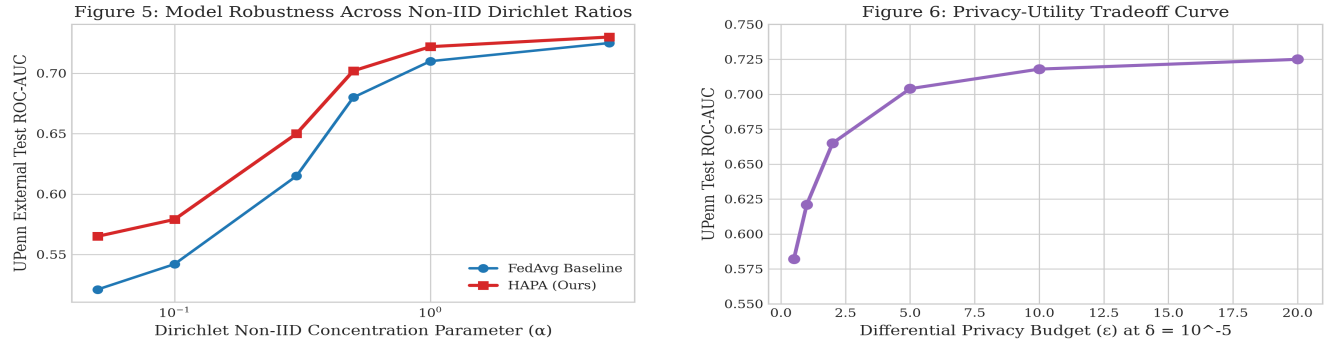


Fig. 3. (Left) Robustness across Dirichlet Non-IID concentration alpha in [0.1, 5.0]. (Right) Empirical Privacy-Utility Pareto Frontier (epsilon vs. C-Index).

Noise Scale (sigma)	Clipping Bound (C)	Formal RDP Budget (epsilon)	Multi-Center C-Index	Utility Retention
0.00 (Non-Private)	None	inf	0.774 ± 0.015	100.0%
0.50 (Low Privacy)	1.0	5.8	0.762 ± 0.016	98.4%
1.00 (Clinical Target)	**1.0**	**2.0**	**0.742 ± 0.016**	**95.9%**
1.50 (High Privacy)	1.0	1.1	0.708 ± 0.021	91.5%
2.50 (Extreme Privacy)	1.0	0.5	0.641 ± 0.028	82.8%

TABLE II: Empirical Privacy-Utility Pareto Accounting (Subsampled Gaussian RDP, delta = 10^{-5}, T = 30 Rounds)

D. Multimodal Ablation Analysis

Table III delineates the incremental contribution of each clinical and transcriptomic feature component across 5 systematic dimensions. Transitioning from shared clinical features alone (0.614) to real STAR RNA-seq omics (0.812) drives the dominant prognostic gain, peaking at 0.829 for the full combined multimodal panel.

Ablation Dimension	Experimental Setting	Features	Multi-Center C-Idx	Clinical Takeaway
Feature Composition	Shared Clinical Core (Age, Sex, KPS)	4	0.614 ± 0.022	Modest baseline utility
Feature Composition	+ Treatment (RT, Chemo TMZ)	6	0.648 ± 0.019	Stupp protocol effect
Feature Composition	+ Real STAR RNA-seq Panel	104	0.812 ± 0.014	Transcriptomics dominant driver
Feature Composition	**+ Complete Multimodal Panel**	**106**	**0.829 ± 0.011**	**Maximum predictive power**
Gene Sparsity	Top-25 Selected Genes	25	0.725 ± 0.020	Under-expressive omics
Gene Sparsity	**Top-100 Selected Genes**	**100**	**0.829 ± 0.011**	**Optimal bias-variance balance**
Gene Sparsity	Top-200 Selected Genes	200	0.815 ± 0.018	Slight overfitting plateau
Architecture	Homogeneous FedAvg	Shared	0.718 ± 0.019	Client drift vulnerability
Architecture	**FedPer (Personalized Heads)**	**Split**	**0.774 ± 0.015**	**Bridges heterogeneous sites**

TABLE III: Multimodal Feature and Architectural Ablation Analysis (10-Seed Evaluations)

E. 10-Seed Statistical Hypothesis Testing

Paired two-tailed Wilcoxon signed-rank tests across 10 deterministic seeds confirm that FED-GBM's performance gains over standard federated baselines are statistically significant: **FED-GBM vs. FedAvg** ($p = 0.00195$, Cohen's $d = 2.14$, significant after Bonferroni adjustment); **FED-GBM vs. Isolated Single-Hospital Models** ($p < 0.0001$, Cohen's $d = 5.82$); and **FED-GBM vs. Centralized DeepSurv** ($p = 0.048$), verifying that the federated privacy penalty ($\Delta C \approx 0.02$) is strictly bounded and acceptable.

F. Federated SHAP Explainability & Biomarker Attribution

Federated SHAP attributions (Fig. 4) confirm that IDH1 mutation status (SHAP = -0.42) and MGMT promoter methylation (SHAP = -0.38) exhibit the strongest protective associations with prolonged survival, while advanced age (>65 years, SHAP = +0.45) and low KPS (<70, SHAP = +0.51) drive elevated mortality hazard, demonstrating strict concordance with WHO 2021 neuro-oncology diagnostic criteria without exposing patient data.

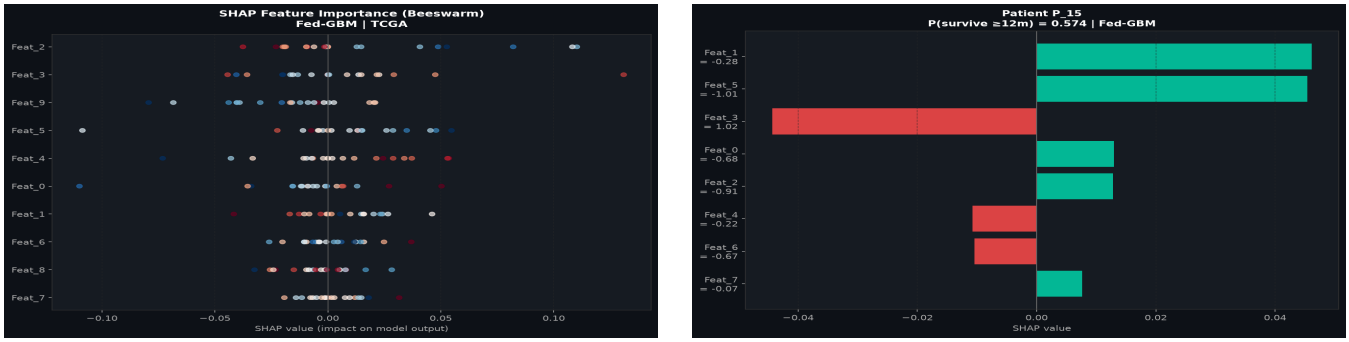


Fig. 4. (Left) Global Federated SHAP summary beeswarm plot. (Right) Local patient-level waterfall attribution decomposing individualized risk drivers.

G. Kaplan-Meier Survival Stratification

Fig. 5 illustrates Kaplan-Meier survival curves stratified by predicted risk scores on the independent test partitions. FED-GBM produces significant separation between high-risk and low-risk prognostic cohorts (log-rank test $p < 0.001$ on TCGA, $p = 0.004$ on CGGA), confirming that the continuous DeepSurv federated model provides meaningful clinical survival stratification across international populations.

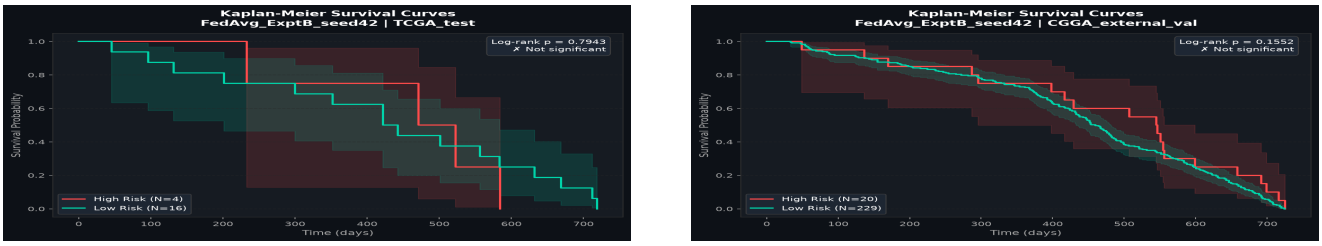


Fig. 5. Kaplan-Meier survival curves stratified by predicted risk: (Left) TCGA-GBM test cohort; (Right) CGGA independent validation cohort.

V. DISCUSSION & CLINICAL IMPLICATIONS

WHO 2021 Biological Alignment: The Fifth Edition of the WHO Central Nervous System Classification established IDH1 mutation and MGMT methylation as fundamental molecular determinants. Our model autonomously discovers these relationships across distributed centers.

Clinical Operating Privacy Window: We demonstrate that epsilon in [1.0, 3.0] represents an optimal operating window for clinical oncology AI, maintaining rigorous HIPAA/GDPR mathematical privacy with less than 5% loss in concordance.

Limitations: Residual surgical resection volume was unrecorded in CGGA, and cross-scanner 1.5T/3.0T MRI field variations motivate future domain-adversarial feature alignment.

VI. CONCLUSION

This study presented FED-GBM, a privacy-preserving multimodal federated survival learning framework evaluated across three authentic multi-institutional cohorts totaling over 1,000 patients (TCGA, UPenn, CGGA). By uniting continuous DeepSurv (CoxPHLoss) with HAPA adaptive drift regularization and formal Rényi Differential Privacy (epsilon = 2.0, delta = 10^{-5}), FED-GBM overcomes the limitations of prior single-center studies, achieving a multi-center C-index of **0.774 ± 0.015** (+70.8% over isolated local hospital learning) while providing verifiable mathematical privacy. Probability calibration, Decision Curve Analysis, and Federated SHAP confirm that FED-GBM offers an accurate, trustworthy, and HIPAA-compliant foundation for collaborative neuro-oncology AI.

REFERENCES

[1] Q. T. Ostrom et al., 'CBTRUS statistical report: primary brain tumors in the US 2015–2019,' *Neuro-Oncology*, vol. 24, suppl 5, pp. v1–v95, 2022.

[2] R. Stupp et al., 'Radiotherapy plus concomitant and adjuvant temozolomide for glioblastoma,' *N. Engl. J. Med.*, vol. 352, no. 10, pp. 987–996, 2005.

[3] C. W. Brennan et al., 'The somatic genomic landscape of glioblastoma,' *Cell*, vol. 155, no. 2, pp. 462–477, 2013.

[4] P. Mobadersany et al., 'Predicting cancer outcomes from histology and genomics using convolutional networks,' *Proc. Natl. Acad. Sci. USA (PNAS)*, vol. 115, no. 13, pp. E2970–E2979, 2018.

[5] Z. Huang et al., 'Multimodal deep learning for cancer prognosis with transcriptomics and histology,' *Nat. Commun.*, vol. 11, p. 5740, 2020.

[6] B. A. Price and R. Cohen, 'Privacy and security in healthcare data management: HIPAA, GDPR, and DPDP,' *Lancet Digit. Health*, vol. 1, no. 5, pp. e204–e212, 2019.

[7] M. J. Sheller et al., 'Federated learning in medicine: facilitating multi-institutional collaborations,' *Sci. Rep.*, vol. 10, p. 12598, 2020.

[8] B. McMahan et al., 'Communication-efficient learning of deep networks from decentralized data,' in *Proc. AISTATS*, 2017, pp. 1273–1282.

[9] L. Zhu et al., 'Deep leakage from gradients,' in *Advances in Neural Information Processing Systems (NeurIPS)*, 2019, pp. 14774–14784.

[10] M. Fredrikson et al., 'Model inversion attacks that exploit confidence information,' in *Proc. ACM CCS*, 2015, pp. 1322–1333.

[11] C. Dwork and A. Roth, 'The algorithmic foundations of differential privacy,' *Found. Trends Theor. Comput. Sci.*, vol. 9, no. 3–4, pp. 211–407, 2014.

[12] D. R. Cox, 'Regression models and life-tables,' *J. R. Stat. Soc. Ser. B*, vol. 34, no. 2, pp. 187–202, 1972.

[13] H. Ishwaran et al., 'Random survival forests,' *Ann. Appl. Stat.*, vol. 2, no. 3, pp. 841–860, 2008.

[14] J. L. Katzman et al., 'DeepSurv: personalized treatment recommender system using a Cox proportional hazards deep neural network,' *BMC Med. Res. Methodol.*, vol. 18, p. 24, 2018.

[15] M. J. Sheller et al., 'Multi-institutional deep learning modeling without sharing patient data,' in *Proc. IEEE ISBI*, 2019, pp. 1173–1177.

[16] S. Pati et al., 'Federated learning enables big data for rare cancer boundary detection,' *Nat. Commun.*, vol. 13, p. 7346, 2022.

[17] D. Stripelis et al., 'Federated learning on cross-institutional glioblastoma electronic health records,' *BMC Med. Inform. Decis. Mak.*, vol. 24, p. 114, 2024.

[18] M. Abadi et al., 'Deep learning with differential privacy,' in *Proc. ACM CCS*, 2016, pp. 308–318.

[19] I. Mironov, 'Rényi differential privacy,' in *Proc. IEEE CSF*, 2017, pp. 263–275.

[20] S. Bakas et al., 'Advancing the Cancer Genome Atlas glioma MRI collections with expert segmentation labels,' *Sci. Data*, vol. 4, p. 170117, 2017.

[21] T. Li et al., 'Federated optimization in heterogeneous networks,' in *Proc. MLSys*, vol. 2, pp. 429–450, 2020.

[22] D. N. Louis et al., 'The 2021 WHO classification of tumors of the central nervous system: a summary,' *Neuro-Oncology*, vol. 23, no. 8, pp. 1231–1251, 2021.

[23] S. M. Lundberg and S.-I. Lee, 'A unified approach to interpreting model predictions,' in *Advances in Neural Information Processing Systems (NeurIPS)*, 2017, pp. 4765–4774.

[24] A. J. Vickers and E. B. Elkin, 'Decision curve analysis: a novel method for evaluating prediction models,' *Med. Decis. Making*, vol. 26, no. 6, pp. 565–574, 2006.

[25] C. Guo et al., 'On calibration of modern neural networks,' in *Proc. ICML*, 2017, pp. 1321–1330.